

MSUG #16

Suunniteltu infra-tiimille, rakennettu kehittäjille
Azure Networking for developers



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Untangling Microsoft Security



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Not Bad Security – an agile specialist in cloud security and data protection, powered by Microsoft capabilities.

Speaker

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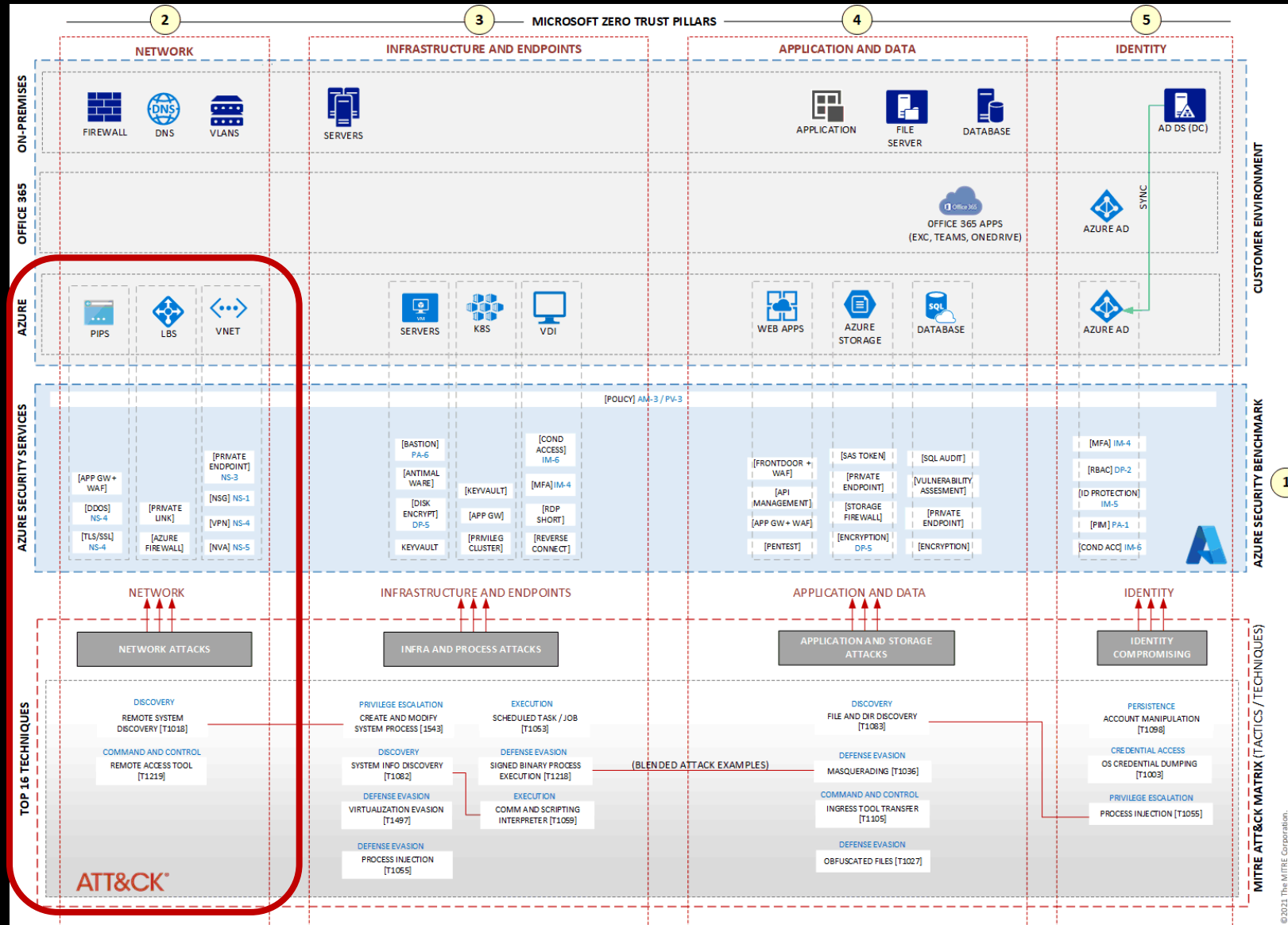
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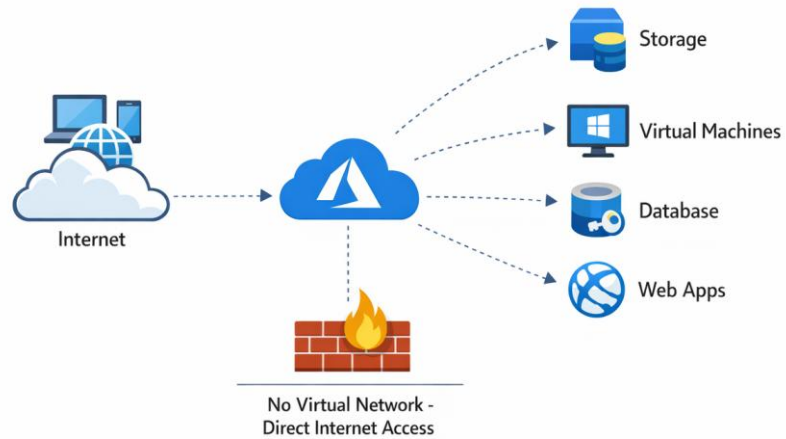


Background & context

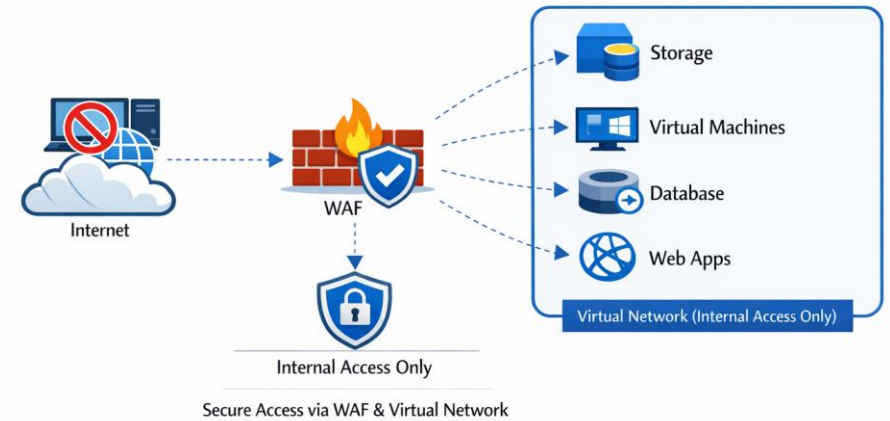


| Background & context

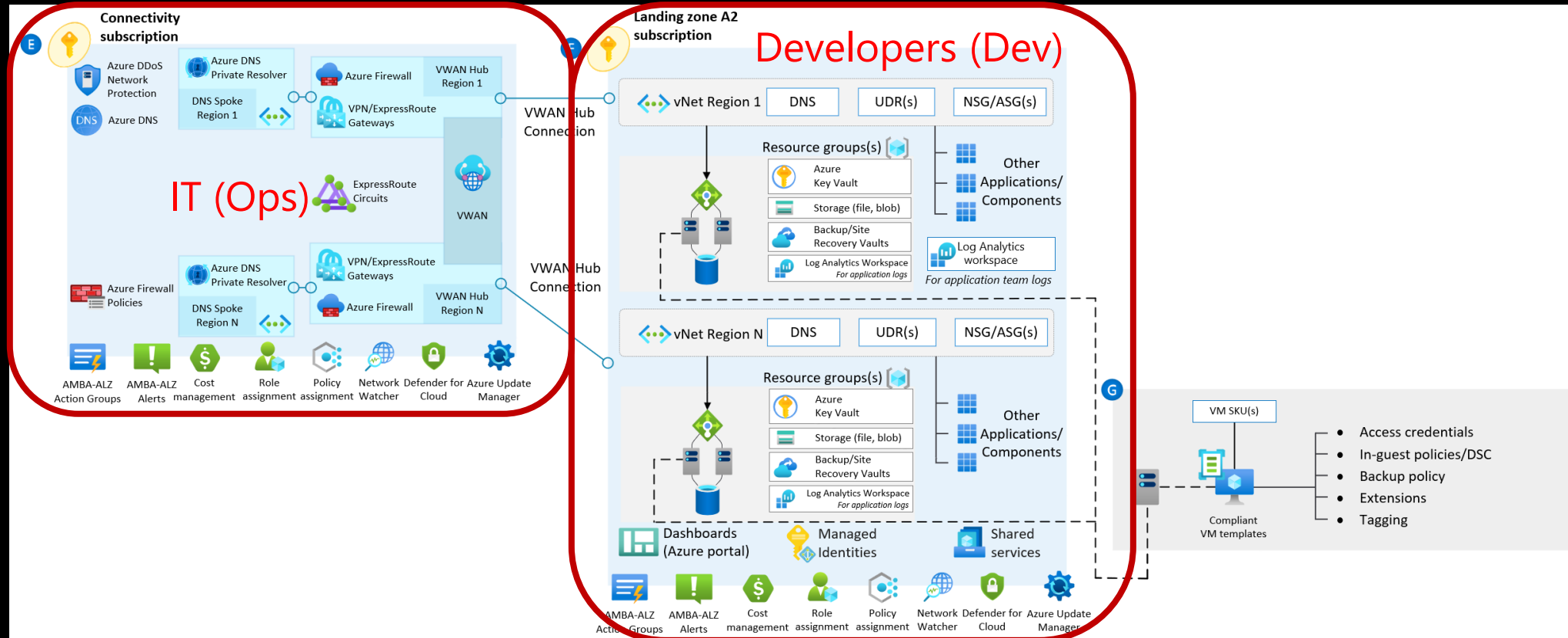
No Virtual Networks: Direct Internet Access



Internal Access Only: Using WAF and Virtual Network



Background & context



| Deployment

Define the policies:

- Resources can't be accessed from internet
- Internet access must be protected with WAF

Policy-driven landing zone:

- Deny policies for your decisions
- Exception process and documentation

Connectivity infrastructure:

- Design & deploy vWAN/traditional hub-n-spoke
- Deploy Front Door/Application Gateway /w WAF

| The Challenge

You incorporate the traditional IT processes in cloud;

Change requests, IT/Ops creating firewall rules, etc. dependencies

You become a bottleneck for developers, and they start hating IT more than usually

You don't build the platform together but instead build it from IT perspective only

You don't communicate, create instructions or educate your developers & cloud users

| (One) Solution

To avoid bottlenecks & dependencies;

Let your developers manage the networking and platform (scary, huh?!)

Your Teams/Slack will be less noisy and developers keep their autonomy

Listen to the developers, software architects and product owners;

This is luckily not too late although would've helped earlier

Create clear ground rules and instructions how to use the platform as a self-service

| Requirements

Your connectivity/platform must be Infrastructure-as-Code and CI/CD driven

Plan your Azure DevOps/GitHub project so it can be “open” for everyone

Infrastructure-as-Code must be modular and easy to use;

No hardcoded modules and parameters in resource definitions

Basic DevOps practices; use branch protection, build validations and pull request reviews

| The New Challenge #1

It's somewhat painful to magically apply the new networking requirements and controls to existing running workloads

Greenfield approach is recommended and almost a necessity often

This will slow down the benefits gained from the new strategy;

Teams need to commit to the cultural & technical changes

Getting to the desired goal is a joint journey but it is often very rewarding and worth it

| The New Challenge #2

Developers have probably never done or understood networking (sorry, generalizing a bit)

Your IT/cloud team might not be familiar with Infrastructure-as-Code either

If the application landing zone level of automation is already high;

You need to solve the dependencies to the platform/connectivity to keep the level of automation high

New changes and testing is required from the existing applications & workloads;

Private endpoints, creating firewall rules to allow what is necessary, etc.

| The New Challenge #3

When public inbound access is disabled;

Access to data planes (key vaults, storage, etc.) doesn't work via internet anymore

You need to provide mechanisms to access the resources from within the virtual network(s)

Jump hosts (VM/PAW), VDI's (Windows 365, AVD), VPN/Express Route, or anything that routes to the target resource/spoke network

This (dramatically) changes how developers need to work going forward

| The New Challenge #4

When public inbound access is disabled;

Existing Microsoft/GitHub hosted agents/runners can't deploy the application anymore ":D"

You need to provide self-hosted agents depending on the DevOps (CI/CD) tools you're using

Brings extra administrative overhead but is manageable

Bonus is that your SDLC security posture will increase and maybe you should've used self-hosted agents already for other reasons

| The New Challenge #5

Congratulations, you now have more problems than before;

But they're a bit different and temporary

Instead of clicking new firewall rules for your developers in Azure portal;

You iterate and educate developers for their own benefit (and review a lot of pull requests)

This is very individual as well – some developers pick up on new things faster than others

.. at least the developers can't complain that IT is so slow and annoying, stopping our projects every day;

Do it yourself now!

| Other Notes from the Field #1

Generally, this changes a lot culturally and removes classical barriers

You are in the Cloud, so why wouldn't you use it as it was designed to

Brings Ops closer to Dev and vice versa

Be there for you developers;

They're going to need support, probably each time when the firewall blocked something – again.

The lead times reduce significantly, and everyone can focus on what's important rather than navigating in the process jungle

| Other Notes from the Field #2

Provide read access to logs and instruct how to troubleshoot networking issues; people will learn

Develop your platform/connectivity Infrastructure-as-Code to be smarter and more automated

Onboarding an existing application to a preventive WAF can be.. ..interesting

When someone asks you; direct them to the documentation instead of helping everyone individually each time – repeating the same thing over and over again

Use AI to support pull request reviewers and developers to help them create secure applications

| Hope you learned something!

